EMCH 574 – Vibration and Waves
Homework 01

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Due date TBD

Contents

Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement reference.
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) reference.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the y-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.1 Analyze Pendulum

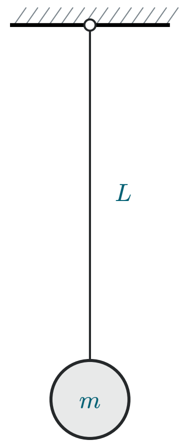


Figure 1: *Pendulum of length L and mass m.*

Consider a pendulum of length L and mass m as shown in Figure 1.

Do the following.

(1) Draw the pendulum displaced by an angle θ from the vertical.

(2) Draw free-body diagram (FBD).

(3) Write Newton's law of motion (NLM) equation.

(4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.

(5) Deduce the governing equation.

(6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation.

(7) Deduce and solve the characteristic equation.

(8) What type are the roots $s_{1,2}$? Choose one of the following options:

- **(a)** Real.
- **(b)** Imaginary.
- **(c)** Complex.

(9) Define the natural frequency ω_n.

(10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n.

(11) Write the solution in terms of exponential functions of $\omega_n t$.

(12) Write the solution in terms of trigonometric functions of $\omega_n t$.

(13) Write the Hz frequency f_n.

(14) Write the period of oscillation τ.

(15) Write the particle motion $u(t)$ in terms of.

- **(a)** Exponential functions.
- **(b)** Trigonometric functions.

(16) Solve the initial value problem (IVP) and find the response $u(t)$ for:

- **(i)** Initial displacement u_0.
- **(ii)** Initial velocity $\dot{u}_0$.
- **(iii)** Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

A.2 Calculate Pendulum Response to Initial Displacement

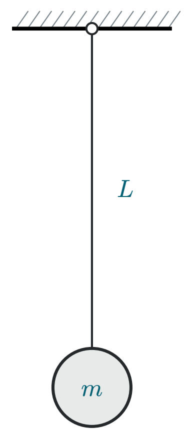


Figure 2: *Pendulum of length L and mass m.*

Take $g = 9.81\,\mathrm{m/s^2}$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565\,\mathrm{mm}$ and mass $m = 52\,\mathrm{kg}$ as shown in Figure 2. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535,\ 562.691,\ 563.845,\ 563.845,\ 565.000,\\ 565.000,\ 566.155,\ 566.155,\ 567.309,\ 568.464]\,\mathrm{mm}.$$

Do the following.

(1) For the ten pendulum sets, calculate.

- **(a)** Natural frequency ω_n in rad/s.
- **(b)** Hz natural frequency f_n.
- **(c)** Period of oscillation τ.
- **(d)** Display in table format L, ω_n, f_n, τ.

(2) Mean values: calculate and display the mean values of ω_n, f_n, τ.

(3) Initial value problem (IVP). Assume initial displacement $u_0 = 2\,\mathrm{mm}$ and zero initial velocity. Use the mean values of ω_n, f_n, and τ to complete items (a)–(c).

- **(a)** Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10\,\mathrm{s}$.
- **(b)** Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 10$.
- **(c)** Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1.

A.3 Calculate Pendulum Response to Impact

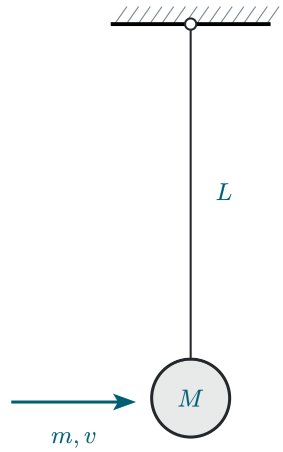


Figure 3: *Impact excitation of a pendulum system.*

Take $g = 9.81\,\mathrm{m/s^2}$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565\,\mathrm{mm}$ and mass $M = 52\,\mathrm{kg}$ as shown in Figure 3. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22\,\mathrm{g}$ and velocity $v = 945\,\mathrm{m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

(1) Display input data.

(2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.

(3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.

(4) Calculate the response $u(t_1)$ after $t_1 = 10\,\mathrm{s}$.

(5) Plot the response of the system for $N_{\mathrm{osc}} = 10$ oscillations and find on the plot:

- **(a)** The largest response.
- **(b)** The response value calculated at item (4).

(6) Discuss your results.

2 Spring-Mass-Damper Vibrating Systems

B.1 Analyze Spring-Mass Vibrating System

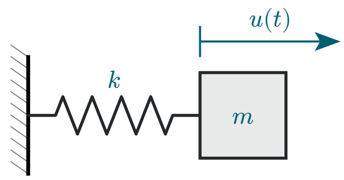


Figure 4: *Spring-mass vibrating system.*

Consider a spring-mass vibrating system as shown in Figure 4.

Do the following.

(1) Draw free-body diagram (FBD).

(2) Write NLM equation.

(3) Deduce the equation of motion (EOM).

(4) Deduce the governing equation. Clearly state your assumptions.

(5) Assume $u(t) = e^{st}$ and substitute into the governing equation.

(6) Deduce the characteristic equation.

(7) Define the natural frequency ω_n in terms of m,k.

(8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n.

(9) What type are the roots $s_{1,2}$? Choose one of the following options:

- **(a)** Real.
- **(b)** Imaginary.
- **(c)** Complex.

(10) Write the solution in terms of exponential functions.

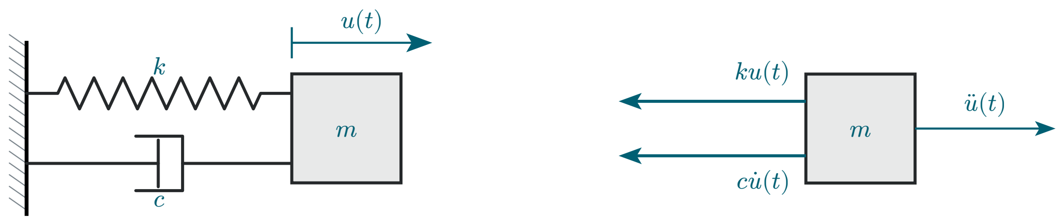
(11) Write the solution in terms of trigonometric functions.

(12) Solve the initial value problem (IVP) and find the response $u(t)$ for:

- **(i)** Initial displacement u_0.
- **(ii)** Initial velocity $\dot{u}_0$.
- **(iii)** Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k, mass, m, and viscous damper, c, as presented in Figure 5. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.



(a) Physical system. (b) Free-body diagram.

Figure 5: *Mass-spring-damper vibrating system.*

Do the following.

(1) State your assumptions in performing this analysis.

(2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.

(3) Write NLM equation.

(4) Deduce the equation of motion (EOM) as an ODE in terms of m, c, k.

(5) Assume $u(t) = e^{st}$ and substitute into EOM.

(6) Deduce the characteristic equation.

(7) Find the roots $s_{1,2}$ of the characteristic equation.

(8) Define critical damping c_{cr} in terms of m and k.

(9) Define damping ratio ζ.

(10) Discuss the values of ζ for the three damping cases:

- **(a)** Underdamped.
- **(b)** Critically damped.
- **(c)** Overdamped.

(11) Express the physical damping c in terms of damping ratio ζ, mass m, and stiffness k.

(12) Define the natural frequency ω_n in terms of m and k.

(13) Express in terms of ω_n, ζ the $s_{1,2}$ roots found at (7).

(14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ.

(15) Define the damped frequency ω_d.

(16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n, ω_d, ζ.

(17) Assume underdamped conditions and state the inequality that ζ should satisfy.

(18) Write the general solution of the ODE deduced at (4) in terms of constants C_1 and C_2 multiplying exponential functions of s_1 and s_2.

(19) Use (16) to write the solution (18) in terms of exponential functions of ω_n, ω_d, ζ.

(20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.

(21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.

(22) Sketch the functions discussed at (21).

(23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity.

(24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

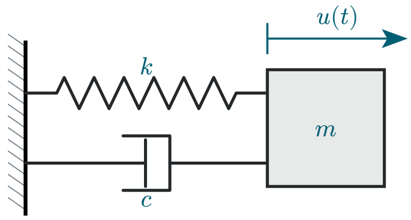


Figure 6: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 12\,\mathrm{kg}$, $k = 655\,\mathrm{N/m}$, $c = 19\,\mathrm{N/(m/s)}$ as shown in Figure 6. The vibrating system is given an initial displacement $u_0 = 3.5\,\mathrm{mm}$ with zero initial velocity.

Do the following.

(1) Display input data.

(2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n.

(3) Calculate the critical damping c_{cr}.

(4) Calculate the damping ratio ζ.

(5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d.

(6) Calculate the damped period of oscillation τ_d.

(7) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2.

(8) Calculate the constants A, B and then the constants C, ϕ.

(9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.

(10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.

(11) Plot the response of the system for $N_{\mathrm{osc}} = 10$ oscillations and use datatips on the plot to find the following:

- **(a)** The largest response.
- **(b)** The response value calculated at item (9).
- **(c)** The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).

(12) Discuss your results.

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k-m-c spring-mass-damper system as shown in Figure 7.

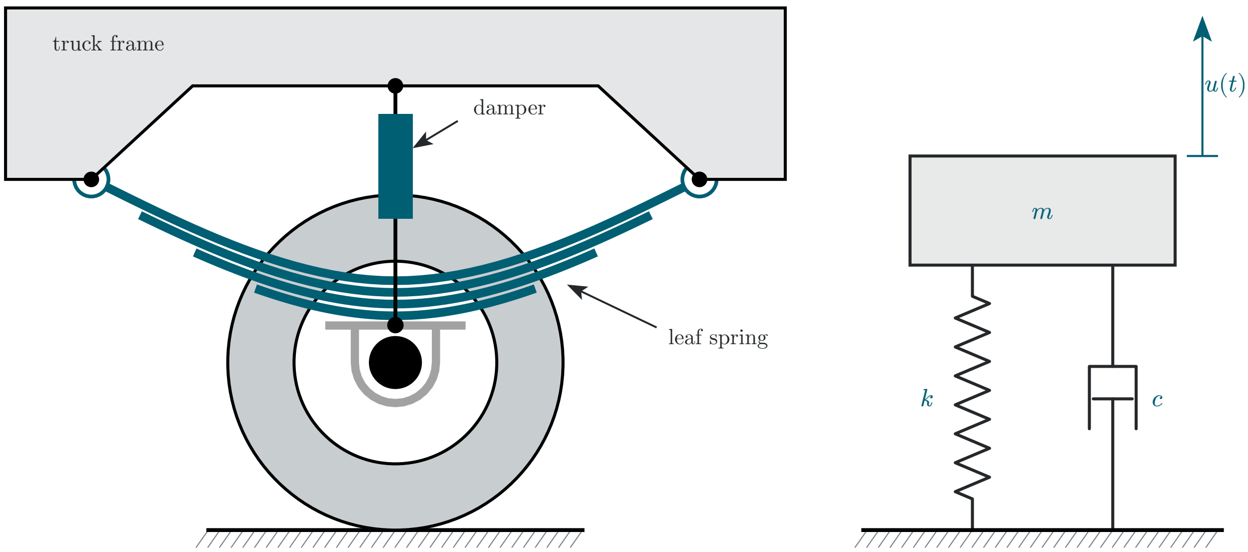


Figure 7: *Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.*

The leaf spring can be approximated by a uniform beam of length $L = 830\,\mathrm{mm}$ and cross-section dimensions $h = 3\,\mathrm{mm}$ and $b = 65\,\mathrm{mm}$, made of steel with $E = 200\,\mathrm{GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 457\,\mathrm{kg}$. Assume initial displacement $u_0 = 125\,\mathrm{mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

(1) Display input data.

(2) Calculate the second moment of area I of the beam cross section.

(3) Calculate the flexural stiffness EI of the beam.

(4) Calculate the spring constant k of the system.

(5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n.

(6) Calculate the critical damping c_{cr}.

(7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c. Print ζ and c.

(8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d.

(9) Calculate the damped period of oscillation τ_d.

(10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2.

(11) Calculate the constants A, B and then the constants C, ϕ.

(12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.

(13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.

(14) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and use datatips on the plot to find the following:

- **(a)** The largest response.
- **(b)** The response value calculated at item (11).
- **(c)** The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).

(15) Discuss your results.

(16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.

- **(a)** Repeat the calculation of items (6) through (12).
- **(b)** Compare and discuss the two results.

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25\,\mathrm{Hz}$ and various values of the damping ratio ζ. The system is given an initial displacement $u_0 = 13\,\mathrm{mm}$ with zero initial velocity.

Do the following.

(1) Display input data.

(2) Calculate the rad/s frequency ω_n.

(3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:

- **(a)** Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ.
- **(b)** Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2.
- **(c)** Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
- **(d)** Take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2, C_1, and C_2.
- **(e)** Plot the response $u(t)$ for $t = 0, \ldots, 3\,\mathrm{s}$.

(4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).

- **(a)** Write the expressions of the two partial solutions $u_1(t)$, $u_2(t)$ in terms of ω_n.
- **(b)** Write the expression of the complete solution in terms of ω_n and C_1, C_2.
- **(c)** Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
- **(d)** Calculate the numerical values of C_1, C_2.
- **(e)** Plot the response $u(t)$ for $t = 0, \ldots, 3\,\mathrm{s}$.

(5) Graduate work. Derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE.

(6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 8.

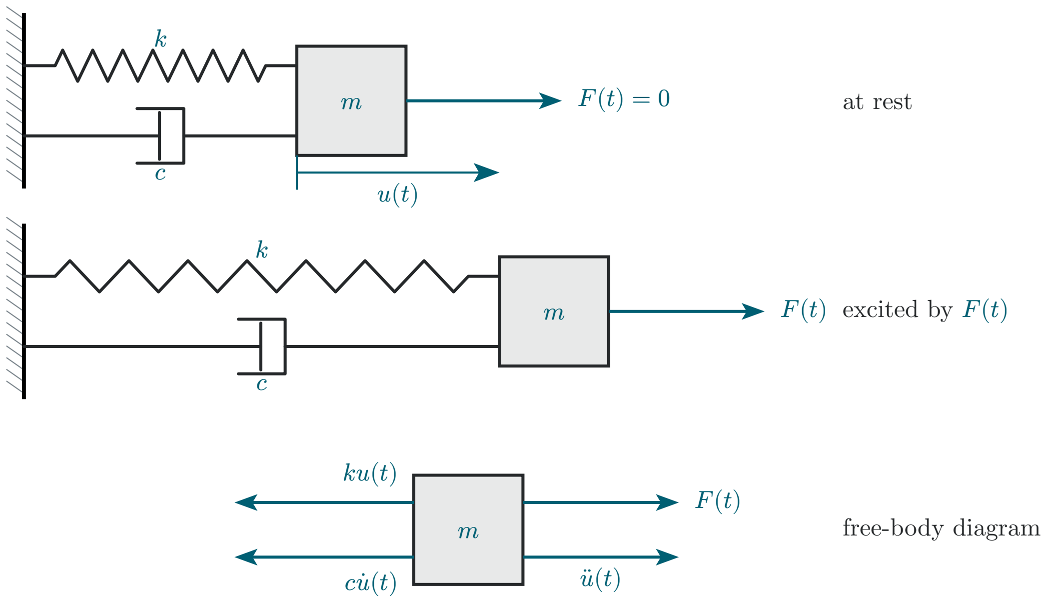


Figure 8: *Spring-mass-damper system excited by force $F(t)$.*

Do the following.

(1) State your assumptions in performing this analysis.

(2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.

(3) Write NLM equation.

(4) Deduce the equation of motion (EOM) as an ODE in terms of m, c, k, $F(t)$.

(5) Assume harmonic excitation $F(t) = \hat{F}\ e^{i\omega t}$, substitute in (4), and divide by m.

(6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n, ζ, $\hat{f}$.

(7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.

(8) Write the general form of the complementary solution $u_c(t)$.

(9) Find the particular solution $u_p(t)$.

(10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.

(11) Find an expression for the steady-state response amplitude $\hat{u}_{\text{ss}}$ in terms of ω_n, ζ, $\hat{f}$, and ω.

(12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n, ζ, ω.

(13) Discuss the variation with frequency of FRF magnitude and phase.

(14) Give the definition of the magnification factor $M(\omega)$.

(15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ.

(16) Give the definition of the resonance frequency ω_r.

(17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ.

(18) Give the expression of the peak magnification M_r.

(19) Give the expression for the FRF phase $\phi(p)$, where $p = \omega/\omega_n$.

(20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.

(21) Calculate the magnification factor at phase resonance.

(22) Calculate the magnitude of the physical response at phase resonance.

(23) Discuss the difference between true resonance and phase resonance.

(24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d.

(25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω.

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

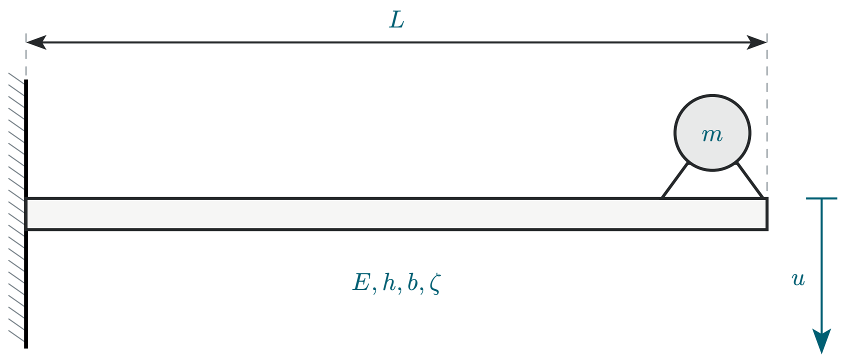


Figure 9: *Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.*

Consider a cantilever beam of length $L = 568\,\text{mm}$, elastic modulus $E = 200\,\text{GPa}$, cross-section dimensions $h = 3\,\text{mm}$ and $b = 37\,\text{mm}$, and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52\,\text{kg}$ operating at $1{,}500\,\text{RPM}$ is attached at the beam tip as shown in Figure 9. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12\,\text{mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

(1) Display input data.

(2) Calculate.

- **(a)** Cross-section second moment of area I, flexural stiffness EI, and spring constant k.
- **(b)** Natural frequency ω_n rad/s, critical damping c_{cr}, physical damping c.
- **(c)** Quasi-static deflection u_{qst} corresponding to $\omega_n \to 0$.

(3) Calculate the values of.

- **(a)** Resonant frequency ω_r and damped frequency ω_d.
- **(b)** Peak magnification M_r.
- **(c)** Magnification at phase resonance M_{90}.
- **(d)** Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
- **(e)** Calculate the Hz frequencies f_r, f_n and the corresponding RPM values.

(4) Assume excitation at phase resonance:

- **(a)** Calculate the excitation frequency ω in rad/s and f in Hz.
- **(b)** Calculate period of oscillation τ.
- **(c)** Plot the time response for $N_{\text{osc}} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.

(5) Graduate work. Complete items (a)–(d).

(a) Calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points.

(b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.

(c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.

(d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact

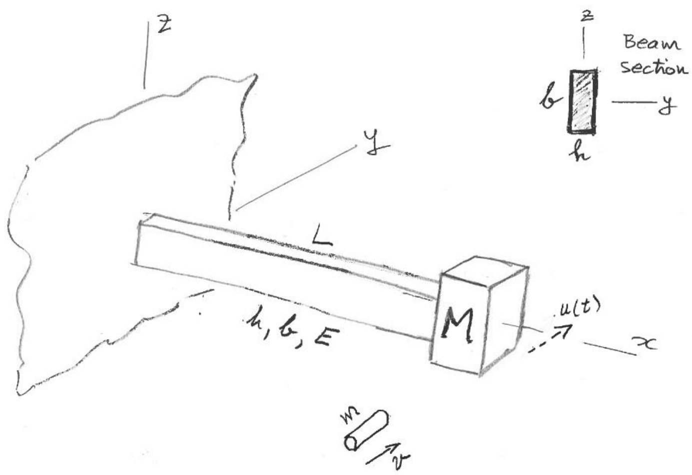


Figure 10: *Impact excitation of a horizontal flex-beam vibration system.*

Consider a cantilever beam of length $L = 568\,\mathrm{mm}$ with a mass $M = 53\,\mathrm{kg}$ attached at its tip as shown in Figure 10. The beam material is steel with $E = 200\,\mathrm{GPa}$. The beam cross-section dimensions are $h = 3\,\mathrm{mm}$, $b = 37\,\mathrm{mm}$. The beam is initially at rest. A small projectile of mass $m = 18\,\mathrm{g}$ and velocity $v = 935\,\mathrm{m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

(1) Display input data.

(2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.

(3) Calculate the cross-sectional stiffness EI of the beam.

(4) Calculate the flex-beam spring constant k.

(5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.

(6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ.

(7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10\,\mathrm{s}$.

(8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:

- **(a)** The largest response.
- **(b)** The response calculated at item (7).

(9) Discuss your results.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

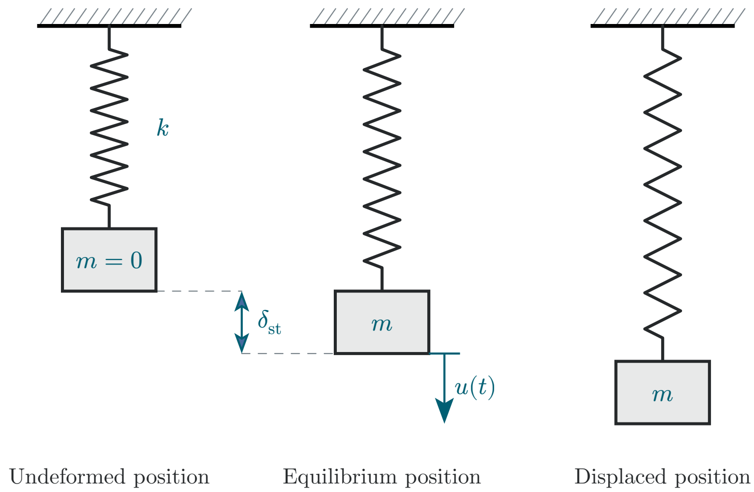


Figure 11: *Vertical spring-mass vibration system.*

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 11.

Do the following.

(1) Assume static equilibrium.

- **(a)** Calculate the weight W acting on the mass m.
- **(b)** Draw free-body diagram (FBD).
- **(c)** Write equilibrium equation.
- **(d)** Solve for the static displacement δ_{st}.

(2) Assume vibration with time-varying vertical displacement $u(t)$.

- **(a)** Draw free-body diagram (FBD).
- **(b)** Deduce equation of motion (EOM).
- **(c)** Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.

(3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.2 Analyze Vertical Flex-Beam Vibration

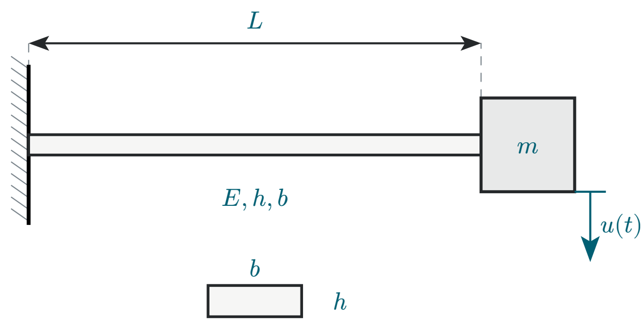


Figure 12: *Vertical flex-beam vibrating system.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 12. The beam cross-section properties are E, h, b.

Do the following.

(1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation.

(2) Calculate the spring constant k of the system.

(3) Assume static equilibrium:

- **(a)** Calculate the weight W acting on the mass m.
- **(b)** Draw free-body diagram (FBD).
- **(c)** Write equilibrium equation.
- **(d)** Solve for the static displacement δ_{st}.

(4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:

- **(a)** Draw free-body diagram (FBD).
- **(b)** Deduce equation of motion (EOM).
- **(c)** Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.

(5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.3 Analyze Horizontal Flex-Beam Vibration

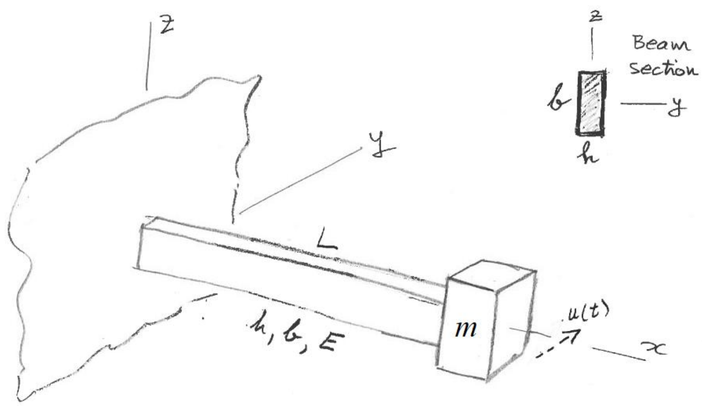


Figure 13: *Horizontal flex-beam vibrating system.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 13. The beam cross-section properties are E, h, b.

Do the following.

(1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 13.

(2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$.

(3) Calculate the flex-beam spring constant k.

(4) Draw free-body diagram (FBD) including all the forces acting on the mass m.

(5) Deduce equation of motion (EOM).

(6) Compare the EOM deduced here with the EOM derived in Problem B.1.

(7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

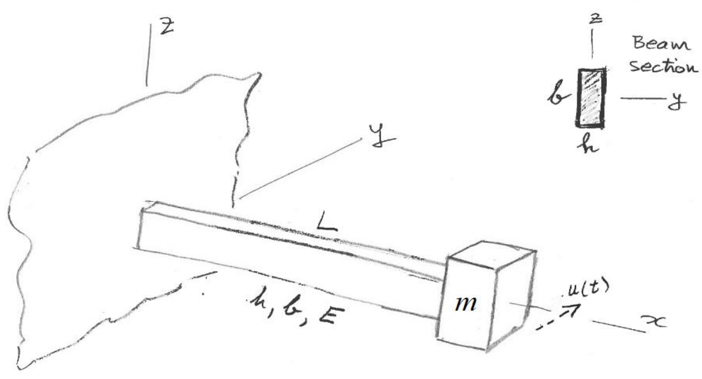


Figure 14: *Flex-beam vibrating system.*

Consider a cantilever beam of length $L = 568\,\mathrm{mm}$ with a mass $m = 5.35\,\mathrm{kg}$ attached at its tip as shown in Figure 14. The beam material is steel with $E = 200\,\mathrm{GPa}$. The beam cross-section dimensions are $h = 3\,\mathrm{mm}$, $b = 37\,\mathrm{mm}$. The tip mass is given an initial displacement $u_0 = 5.4\,\mathrm{mm}$ with zero initial velocity.

Do the following.

(1) Display input data.

(2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.

(3) Calculate the cross-sectional stiffness EI of the beam.

(4) Calculate the spring constant k of the system.

(5) Calculate the natural frequency in rad/sec ω_n.

(6) Calculate the natural frequency in Hz f_n.

(7) Calculate the period of oscillation in sec τ.

(8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10\,\mathrm{s}$.

(9) Plot the response of the system for $N_{\mathrm{osc}} = 25$ oscillations and use a data tip to find $u(t_1)$ at t_1.

(10) Discuss your results.

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 15.

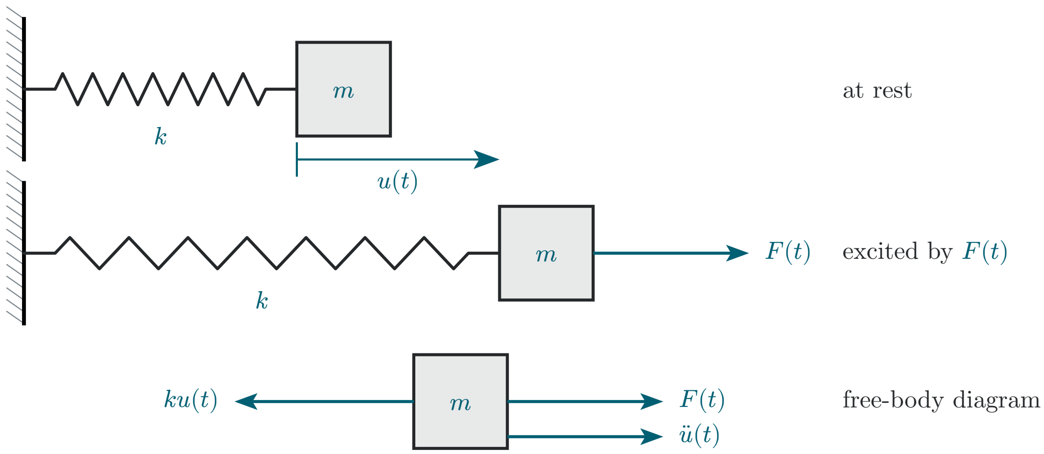


Figure 15: *Spring-mass system excited by force $F(t)$.*

General Analysis of Spring-Mass Forced Vibration

Do the following.

(1) State your assumptions in performing this analysis.

(2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.

(3) Write NLM equation.

(4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m, c, k, $F(t)$.

(5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.

(6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation studied in HW01 and write it in terms of trigonometric functions multiplied by constants A and B.

Spring-Mass Response to a Suddenly Applied Load

(7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0.

(8) Find the particular solution for this case of a suddenly applied load.

(9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8).

(10) Assume "at rest" initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0, k, and $\cos\omega_n t$.

(11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st}.

C.6 Calculate Cantilever Response to Suddenly Applied Weight

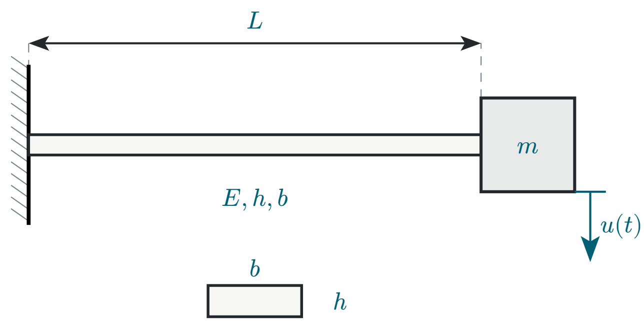


Figure 16: *Cantilever beam with tip mass.*

Take $g = 9.81\,\mathrm{m/s^2}$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568\,\mathrm{mm}$ and a mass $m = 5.35\,\mathrm{kg}$ that has to be attached to its tip as shown in Figure 16. The beam is made of steel with elastic modulus $E = 200\,\mathrm{GPa}$, yield stress $Y = 710\,\mathrm{MPa}$, and cross-section dimensions $h = 3\,\mathrm{mm}$ and $b = 37\,\mathrm{mm}$.

Do the following.

(1) Display input data.

(2) Calculate the cross-section second moment of area I, the flexural stiffness EI, the spring constant k, the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.

(3) Calculate the weight F_0 of the mass m.

(4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st}.

(5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.

- **(a)** Calculate the maximum dynamic displacement u_{dyn}.
- **(b)** Plot the response $u(t)$ measured from the undeformed position for $N_{\mathrm{osc}} = 10$ oscillations.
- **(c)** Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).

(6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $\mathrm{SF} = Y/\sigma$, and comment on your results for the following two cases.

- **(a)** Static-case stress σ_{st}.
- **(b)** Dynamic-case stress σ_{dyn}.

(7) Discuss your results.

C.7 Analyze Flex-Beam Pendulum

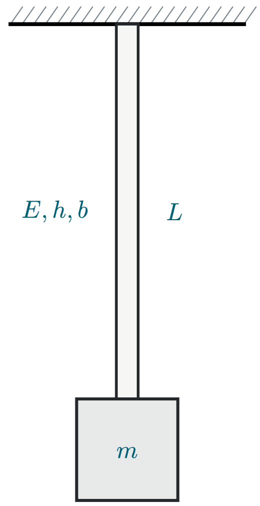


Figure 17: *Flex-beam pendulum.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 17. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b.

Do the following.

(1) Calculate the minor and major second moments of area I_1 and I_2.

(2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1.

(3) Draw free-body diagram (FBD).

(4) Write Newton's law of motion (NLM) equation.

(5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.

(6) Define the natural frequency $\omega_{n,\mathrm{FBP}}$ of the flex-beam pendulum.

(7) Deduce the governing equation.

(8) Compare your results with those of Problem B.1 and write the general solution.

(9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

(10) Graduate work. Explain how the solution would change if flexure were about I_2.

C.8 Calculate Flex-Beam Pendulum Response to Impact

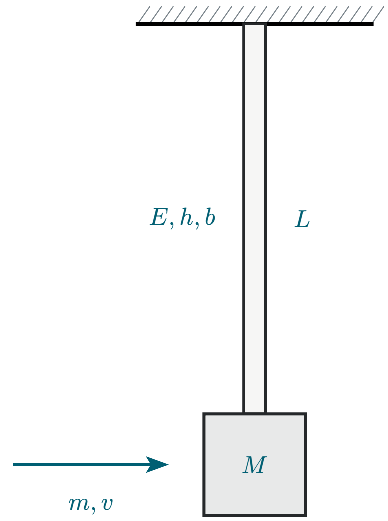


Figure 18: *Impact excitation of a vertical flex-beam pendulum.*

Take $g = 9.81\,\mathrm{m/s^2}$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568\,\mathrm{mm}$ with a mass $M = 42\,\mathrm{kg}$ attached at its end, as shown in Figure 18. The beam is steel with $E = 200\,\mathrm{GPa}$ and cross-section dimensions $h = 3\,\mathrm{mm}$ and $b = 37\,\mathrm{mm}$. The beam is initially at rest. A projectile of mass $m = 18\,\mathrm{g}$ and velocity $v = 935\,\mathrm{m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

(1) Display input data.

(2) Calculate the minor second moment of area I of the beam.

(3) Calculate the minor cross-sectional stiffness EI.

(4) Calculate the flex-beam spring constant k.

(5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.

(6) Calculate the rad/s natural frequency $\omega_{n,\mathrm{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.

(7) Calculate the response $u(t_1)$ after $t_1 = 10\,\mathrm{s}$.

(8) Plot the response of the system for $N_{\mathrm{osc}} = 10$ oscillations and find on the plot:

- **(a)** The largest response.
- **(b)** The response value calculated at item (7).

(9) Compare your results with those of Problem A.3.

C.9 Complex Exponentials and Trigonometric Functions

(1) Find the solutions of $z^4 = 1$ (the fourth roots of unity) and express them as.

(a) Complex numbers.

(b) Complex exponentials.

(2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula).

(3) Calculate the phase difference ϕ in radians and degrees between the following signals.

(a) Compare $x_1(t) = \sin(\omega t)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.

(b) Compare $x_1(t) = \sin(\omega t + \pi/3)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.

(c) Compare $x_1(t) = \sin(\omega t + 45°)$ and $x_2(t) = \sin(\omega t + 115°)$.

(4) What is the phase in radians and degrees of the following complex numbers:

(a) $e^{i\pi/2}$.

(b) $e^{i3\pi/2}$.

(c) $e^{i\pi/4}$.

(d) $e^{-i\pi/2}$.

(e) $e^{-i\pi/4}$.

(5) Draw the locations on the unit circle of the complex numbers in Section (3) above.

(6) Calculate the magnitude and phase in degrees of the complex number $Z = 4 + 5i$.